

LeafAlias: real-time importance sampling of dynamic emissive textures for video-based lighting

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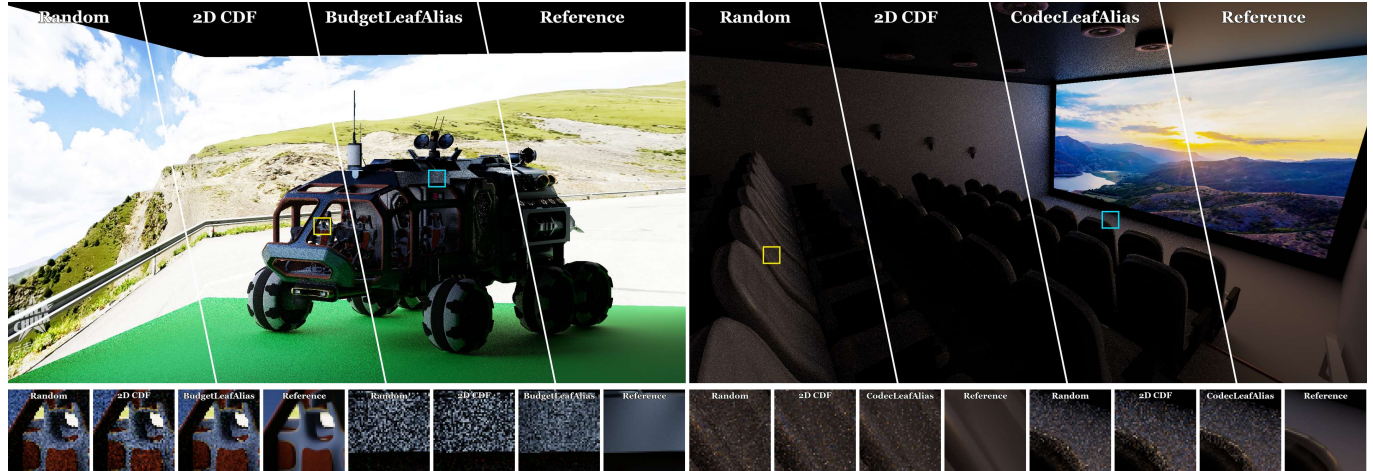


Fig. 1. Two scenes lit by a single dynamic, video-driven emissive texture, rendered with our LeafAlias importance sampler. (Left) The Rover scene — a virtual-production LED wall displaying a looping road/mountain video panorama that lights the vehicle directly, sampled here with BudgetLeafAlias. (Right) The Cinema scene — a theater lit only by the animated sunset footage playing on its own screen, sampled with CodecLeafAlias. Each panel compares, left to right: Random sampling, 2D CDF inversion, our technique, and a converged reference, all at their calibrated real-time budgets (≈ 33 ms).

We present a method for fast importance sampling HDR textured lights that allows real-time video-driven lighting. Our method partitions a light's texture into a budgeted, variance-driven quadtree and builds an alias table over the resulting leaves, avoiding the cost of a dense per-frame CDF while still capturing the texture's overall energy distribution. We present two variants: one that reuses the per-frame hierarchical partitions already present in the metadata of modern video codecs, and another that builds a budgeted quadtree to fit the entire sampling structure in the L1 cache of the GPU. Compared to the dense 2D CDF-inversion baseline, at a matched real-time frame-time budget our technique reduces RMSE by 9–53% depending on scene, by letting the saved sampling cost be reinvested in additional light samples. It also reduces memory consumption and transfers by several orders of magnitude when compared to state-of-the-art samplers.

CCS Concepts: • **Computing methodologies** → **Image-based rendering**; **Image compression**; **Ray tracing**.

Additional Key Words and Phrases: Image-based lighting, video codecs, Alias tables, emissive textures, real-time rendering

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1 Introduction

Importance sampling of emissive textures is essential for efficient image-based lighting, and increasingly critical in production and VFX workflows where video is used directly as a dynamic light source. A single emissive texture lets a renderer reproduce the directional detail of a real environment far beyond what analytic lights can approximate. Traditionally, this texture has been static: captured once and reused for an entire sequence. However, there are recent use cases such as LED-wall virtual production, AR/VR object insertion, and video-based environment maps that require illumination that changes every frame.

State-of-the-art texture light sampling techniques are designed for static images. They rely on offline construction of large sampling structures that are stored and reused. The most common example of this strategy is how rendering and ray tracing toolboxes deal with HDR environment maps. The standard approach is to build a two-dimensional piecewise-constant distribution from the texture's luminance: a marginal CDF over rows and a per-row conditional CDF. The sampler is then inverted via binary search at sample time, which is expensive and produces incoherent memory access. A more recent and well-adopted approach is to generate a mipmap-style hierarchical importance map [Clarberg et al. 2005]. At sample time, a hierarchical descent through a mip pyramid of the texture